Coil Sensitivity and Perturbation Analysis

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1 Background: Magnetic Field Modeling

Consider an array of 4×4 electromagnetic sub-coils. Each sub-coil (i, j) is located at position $\mathbf{p}_{i,j} \in \mathbb{R}^3$ with orientation unit vector $\mathbf{n}_{i,j} \in \mathbb{S}^2$, as depicted in Figure **??**. Each coil is driven by an AC current $I_{i,j}(t)$, defined by:

$$I_{i,j}(t) = I_0 \cos(\omega t + \psi_{i,j}), \tag{1}$$

where I_0 is the peak AC current amplitude, ω is the drive angular frequency, and $\psi_{i,j}$ is the relative actuation phase shift of coil (i, j).

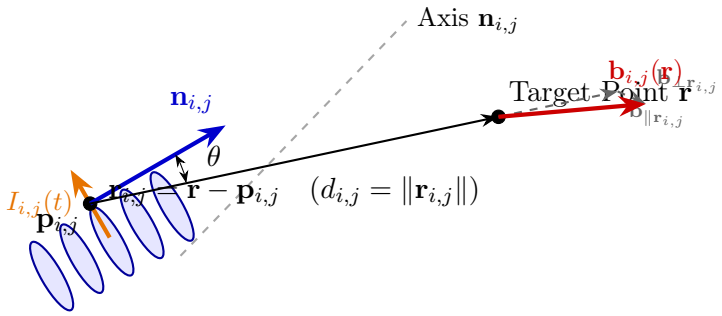


Figure 1: Geometry of a single sub-coil (i, j) located at position $\mathbf{p}_{i,j}$ with orientation unit vector $\mathbf{n}_{i,j}$ driven by time-varying current $I_{i,j}(t)$, generating magnetic field response $\mathbf{b}_{i,j}(\mathbf{r})$ at target point $\mathbf{r}$.

Under the point-dipole approximation, the magnetic flux density response per unit current generated by sub-coil (i, j) at target point $\mathbf{r} \in \mathbb{R}^3$ is given by:

$$\mathbf{b}_{i,j}(\mathbf{r}) = K_{\text{dipole}} \left(\frac{3(\mathbf{r}_{i,j} \cdot \mathbf{n}_{i,j})\mathbf{r}_{i,j}}{d_{i,j}^5} - \frac{\mathbf{n}_{i,j}}{d_{i,j}^3} \right), \tag{2}$$

where $\mathbf{r}_{i,j} = \mathbf{r} - \mathbf{p}_{i,j}$ is the relative position vector, $d_{i,j} = \|\mathbf{r}_{i,j}\|$ is the Euclidean distance, and $K_{\text{dipole}} = \frac{\mu_0 m_0}{4\pi}$ represents the unit-current magnetic dipole strength prefactor ($m_0 = NA$ for a coil with N turns and effective cross-sectional area A).

Applying linear superposition across all 16 driven coils and expanding $I_{i,j}(t)$ using the cosine angle addition identity gives the total time-varying field $\mathbf{B}(\mathbf{r}, t)$:

$$\mathbf{B}(\mathbf{r}, t) = \sum_{i=1}^{4} \sum_{j=1}^{4} \mathbf{b}_{i,j}(\mathbf{r}) I_{i,j}(t) = \mathbf{u}(\mathbf{r}) \cos(\omega t) + \mathbf{v}(\mathbf{r}) \sin(\omega t), \tag{3}$$

where the spatial field is resolved into orthogonal in-phase $\mathbf{u}(\mathbf{r})$ and 90°-lagged (quadrature) $\mathbf{v}(\mathbf{r})$ vector components:

$$\mathbf{u}(\mathbf{r}) = I_0 \sum_{i=1}^{4} \sum_{j=1}^{4} \mathbf{b}_{i,j}(\mathbf{r}) \cos \psi_{i,j}, \qquad \mathbf{v}(\mathbf{r}) = -I_0 \sum_{i=1}^{4} \sum_{j=1}^{4} \mathbf{b}_{i,j}(\mathbf{r}) \sin \psi_{i,j}. \tag{4}$$

From $\mathbf{u}(\mathbf{r})$ and $\mathbf{v}(\mathbf{r})$, the field rotation vector $\mathbf{w}(\mathbf{r})$, local rotation axis $\mathbf{n}_{\text{rot}}(\mathbf{r})$, root-mean-square (RMS) field magnitude $B_{\text{rms}}(\mathbf{r})$, and lateral magnetic force driver gradient $\mathbf{g}_{\text{lat}}(\mathbf{r})$ are computed as:

$$\mathbf{w}(\mathbf{r}) = \mathbf{u}(\mathbf{r}) \times \mathbf{v}(\mathbf{r}), \qquad \mathbf{n}_{\text{rot}}(\mathbf{r}) = \frac{\mathbf{w}(\mathbf{r})}{\|\mathbf{w}(\mathbf{r})\|}, \tag{5}$$

$$B_{\text{rms}}(\mathbf{r}) = \sqrt{\frac{1}{2}\left(\|\mathbf{u}(\mathbf{r})\|^2 + \|\mathbf{v}(\mathbf{r})\|^2\right)}, \qquad \mathbf{g}_{\text{lat}}(\mathbf{r}) = \nabla_{xy}\left(B_{\text{rms}}^2(\mathbf{r})\right). \tag{6}$$

2 Frames and Kinematics in $\mathfrak{se}(3)$

We define frames and sub-coil poses using homogeneous transformation matrices $T_{i,j} \in SE(3)$, parameterized by rotation matrix $R_{i,j} \in SO(3)$ (where $\mathbf{n}_{i,j} = R_{i,j}\mathbf{e}_z$) and translation vector $\mathbf{p}_{i,j} \in \mathbb{R}^3$:

$$T_{i,j} = \begin{bmatrix} R_{i,j} & \mathbf{p}_{i,j} \\ \mathbf{0}^T & 1 \end{bmatrix}. \tag{7}$$

When performing operations across frame transformations, matrix compositions combine both components directly:

$$T_{W \leftarrow C_i} = \text{Rot}_Z(\Phi_i)\, T_{W \leftarrow C_1}, \qquad \mathbf{p}_{i,j} = R_i \mathbf{p}_j^{(t)}, \qquad \mathbf{n}_{i,j} = R_i \mathbf{n}_j^{(t)}. \tag{8}$$

3 Additive Gaussian Pose Noise

For each coil pose $T_{i,j}$, we apply additive zero-mean Gaussian noise $\boldsymbol{\xi}_{i,j} = [\boldsymbol{\omega}_{i,j}^T, \boldsymbol{\nu}_{i,j}^T]^T \in \mathfrak{se}(3)$:

$$\boldsymbol{\xi}_{i,j} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\xi), \qquad \boldsymbol{\Sigma}_\xi = \text{diag}(\sigma_{\text{rot}}^2 \mathbf{I}_3, \sigma_{\text{trans}}^2 \mathbf{I}_3). \tag{9}$$

To first order, position perturbations $\delta\mathbf{p}_{i,j}$ and orientation perturbations $\delta\mathbf{n}_{i,j}$ follow:

$$\delta\mathbf{p}_{i,j} = \boldsymbol{\nu}_{i,j}, \qquad \delta\mathbf{n}_{i,j} = \boldsymbol{\omega}_{i,j} \times \mathbf{n}_{i,j} = -\hat{\mathbf{n}}_{i,j}\boldsymbol{\omega}_{i,j}, \tag{10}$$

where $\hat{\mathbf{a}}$ denotes the 3×3 skew-symmetric cross-product matrix of vector $\mathbf{a}$:

$$\hat{\mathbf{a}} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}. \tag{11}$$

4 Objective Function Formulation

We formulate general cost terms integrated over a 2D/3D region of interest Ω:

$$\mathcal{J}_{\text{sens}} = \int_\Omega \sum_{i=1}^{4}\sum_{j=1}^{4} \text{Tr}\left(\mathbf{J}_{i,j}^{\mathbf{n}_{\text{rot}}}(\mathbf{r}) \boldsymbol{\Sigma}_\xi \left(\mathbf{J}_{i,j}^{\mathbf{n}_{\text{rot}}}(\mathbf{r})\right)^T\right) d\mathbf{r}, \tag{12}$$

$$\mathcal{J}_{\text{mag}} = -\int_\Omega B_{\text{rms}}(\mathbf{r})\, d\mathbf{r}, \qquad \mathcal{J}_{\text{grad}} = \int_\Omega \|\mathbf{g}_{\text{lat}}(\mathbf{r})\|^2\, d\mathbf{r}. \tag{13}$$

We investigate coupled coil plates versus individually movable coil designs, modeling perturbations as follows:

- **Coupled Coil Plates:** Mechanical noise cascades through the assembly. Variance is added at the cluster baseplate level ($\boldsymbol{\Sigma}_{\text{plate}}$) as well as the individual sub-coil mounts ($\boldsymbol{\Sigma}_{\text{coil}}$), doubling total variance ($\boldsymbol{\Sigma}_\xi = \boldsymbol{\Sigma}_{\text{plate}} + \boldsymbol{\Sigma}_{\text{coil}} \approx 2\boldsymbol{\Sigma}_{\text{coil}}$).
- **Individually Mounted Coils:** Each sub-coil is positioned independently, resulting in single-level noise ($\boldsymbol{\Sigma}_\xi = \boldsymbol{\Sigma}_{\text{coil}}$).

5 Perturbation Propagation

To implement this sensitivity framework in symbolic optimization tools like CasADi, we physically derive the exact closed-form Jacobians mapping $\mathfrak{se}(3)$ perturbations $\boldsymbol{\xi}_{i,j}$ to the rotation axis variation $\delta\mathbf{n}_{\text{rot}}(\mathbf{r})$.

5.1 Physical Derivation of Single-Coil Flux Density Sensitivity

Consider the dipole response $\mathbf{b}_{i,j}(\mathbf{r})$. Taking the total differential with respect to position vector $\mathbf{p}_{i,j}$ and unit normal vector $\mathbf{n}_{i,j}$ gives:

$$\delta\mathbf{b}_{i,j} = \frac{\partial\mathbf{b}_{i,j}}{\partial\mathbf{p}_{i,j}}\delta\mathbf{p}_{i,j} + \frac{\partial\mathbf{b}_{i,j}}{\partial\mathbf{n}_{i,j}}\delta\mathbf{n}_{i,j}. \tag{14}$$

1. Position Derivative Matrix ($\mathbf{J}^{\mathbf{b}}_{\mathbf{p}_{i,j}} \in \mathbb{R}^{3\times3}$): Since $\mathbf{r}_{i,j} = \mathbf{r} - \mathbf{p}_{i,j}$, taking spatial gradients with respect to $\mathbf{p}_{i,j}$ yields $\frac{\partial\mathbf{r}_{i,j}}{\partial\mathbf{p}_{i,j}} = -\mathbf{I}_3$ and $\frac{\partial d_{i,j}}{\partial\mathbf{p}_{i,j}} = -\frac{\mathbf{r}^T_{i,j}}{d_{i,j}}$. Differentiating $\mathbf{b}_{i,j}$ using product and chain rules yields:

$$\mathbf{J}^{\mathbf{b}}_{\mathbf{p}_{i,j}} = \frac{\partial\mathbf{b}_{i,j}}{\partial\mathbf{p}_{i,j}} = -K_{\text{dipole}}\left[\frac{3(\mathbf{r}^T_{i,j}\mathbf{n}_{i,j})\mathbf{I}_3 + 3\mathbf{r}_{i,j}\mathbf{n}^T_{i,j} + 3\mathbf{n}_{i,j}\mathbf{r}^T_{i,j}}{d^5_{i,j}} - \frac{15(\mathbf{r}^T_{i,j}\mathbf{n}_{i,j})\mathbf{r}_{i,j}\mathbf{r}^T_{i,j}}{d^7_{i,j}}\right]. \tag{15}$$

- $d^{-5}_{i,j}$: Accounts for second-order spatial decay of the magnetic gradient tensor.
- $d^{-7}_{i,j}$: Captures higher-order geometric dilation from distance perturbations along vector $\mathbf{r}_{i,j}$.

2. Orientation Derivative Matrix ($\mathbf{J}^{\mathbf{b}}_{\boldsymbol{\omega}_{i,j}} \in \mathbb{R}^{3\times3}$): Differentiating $\mathbf{b}_{i,j}$ with respect to coil normal vector $\mathbf{n}_{i,j}$ gives:

$$\frac{\partial\mathbf{b}_{i,j}}{\partial\mathbf{n}_{i,j}} = K_{\text{dipole}}\left[\frac{3\mathbf{r}_{i,j}\mathbf{r}^T_{i,j}}{d^5_{i,j}} - \frac{\mathbf{I}_3}{d^3_{i,j}}\right]. \tag{16}$$

Substituting the differential rotation kinematic relationship $\delta\mathbf{n}_{i,j} = -\hat{\mathbf{n}}_{i,j}\boldsymbol{\omega}_{i,j}$ into the total differential yields:

$$\mathbf{J}^{\mathbf{b}}_{\boldsymbol{\omega}_{i,j}} = \frac{\partial\mathbf{b}_{i,j}}{\partial\mathbf{n}_{i,j}}\frac{\partial\mathbf{n}_{i,j}}{\partial\boldsymbol{\omega}_{i,j}} = -K_{\text{dipole}}\left[\frac{3\mathbf{r}_{i,j}\mathbf{r}^T_{i,j}}{d^5_{i,j}} - \frac{\mathbf{I}_3}{d^3_{i,j}}\right]\hat{\mathbf{n}}_{i,j}. \tag{17}$$

Concatenating both blocks forms the block flux density Jacobian $\mathbf{J}^{\mathbf{b}}_{i,j} = [\mathbf{J}^{\mathbf{b}}_{\boldsymbol{\omega}_{i,j}} \;\; \mathbf{J}^{\mathbf{b}}_{\mathbf{p}_{i,j}}] \in \mathbb{R}^{3\times6}$ such that $\delta\mathbf{b}_{i,j} = \mathbf{J}^{\mathbf{b}}_{i,j}\boldsymbol{\xi}_{i,j}$.

5.2 Propagation to Time-Separated Field Vectors

Because the total field is a linear combination of individual sub-coil fields weighted by current phase $\psi_{i,j}$, the variations in in-phase vector $\mathbf{u}(\mathbf{r})$ and lagging vector $\mathbf{v}(\mathbf{r})$ with respect to pose noise $\boldsymbol{\xi}_{i,j}$ are:

$$\frac{\partial\mathbf{u}}{\partial\boldsymbol{\xi}_{i,j}} = I_0\cos\psi_{i,j}\mathbf{J}^{\mathbf{b}}_{i,j}, \qquad \frac{\partial\mathbf{v}}{\partial\boldsymbol{\xi}_{i,j}} = -I_0\sin\psi_{i,j}\mathbf{J}^{\mathbf{b}}_{i,j}. \tag{18}$$

5.3 Cross-Product Propagation to Rotation Vector $\mathbf{w}(\mathbf{r})$

The unnormalized rotation vector is defined as $\mathbf{w}(\mathbf{r}) = \mathbf{u}(\mathbf{r}) \times \mathbf{v}(\mathbf{r})$. Differentiating via the vector product rule gives:

$$\delta\mathbf{w} = \delta\mathbf{u}\times\mathbf{v} + \mathbf{u}\times\delta\mathbf{v} = -\mathbf{v}\times\delta\mathbf{u} + \mathbf{u}\times\delta\mathbf{v}. \tag{19}$$

Converting cross products to skew-symmetric matrix multiplication ($-\mathbf{v}\times\delta\mathbf{u} = -\hat{\mathbf{V}}\delta\mathbf{u}$ and $\mathbf{u}\times\delta\mathbf{v} = \hat{\mathbf{U}}\delta\mathbf{v}$) yields the Jacobian $\mathbf{J}^{\mathbf{w}}_{i,j} = \frac{\partial\mathbf{w}}{\partial\boldsymbol{\xi}_{i,j}} \in \mathbb{R}^{3\times6}$:

$$\mathbf{J}^{\mathbf{w}}_{i,j} = -\hat{\mathbf{V}}\left(I_0\cos\psi_{i,j}\mathbf{J}^{\mathbf{b}}_{i,j}\right) - \hat{\mathbf{U}}\left(I_0\sin\psi_{i,j}\mathbf{J}^{\mathbf{b}}_{i,j}\right). \;\Big) \qquad (209701\ \text{Tf}\ 13.3$$

5.4 Projection to Unit Rotation Axis $\mathbf{n_{rot}}(\mathbf{r})$

Differentiating the normalization operator $\mathbf{n}_{\text{rot}} = \frac{\mathbf{w}}{\|\mathbf{w}\|}$ with respect to vector $\mathbf{w}$ yields the orthogonal projection matrix $\mathbf{P}_{\mathbf{w}}^{\perp}$:

$$\frac{\partial \mathbf{n}_{\text{rot}}}{\partial \mathbf{w}} = \frac{1}{\|\mathbf{w}\|}\left(\mathbf{I}_3 - \frac{\mathbf{w}\mathbf{w}^T}{\|\mathbf{w}\|^2}\right) = \frac{1}{\|\mathbf{w}\|}\left(\mathbf{I}_3 - \mathbf{n}_{\text{rot}}\mathbf{n}_{\text{rot}}^T\right). \tag{21}$$

Applying the chain rule gives the final composite Jacobian $\mathbf{J}_{i,j}^{\mathbf{n}_{\text{rot}}} = \frac{\partial \mathbf{n}_{\text{rot}}}{\partial \boldsymbol{\xi}_{i,j}} \in \mathbb{R}^{3\times 6}$:

$$\mathbf{J}_{i,j}^{\mathbf{n}_{\text{rot}}} = \frac{1}{\|\mathbf{w}\|}\left(\mathbf{I}_3 - \mathbf{n}_{\text{rot}}\mathbf{n}_{\text{rot}}^T\right)\mathbf{J}_{i,j}^{\mathbf{w}}. \tag{22}$$

This closed-form formulation matches the graph-evaluation structure implemented in CasADi symbolic graphs, enabling rapid parallel evaluation of $\mathcal{J}_{\text{sens}}$ without Monte Carlo sampling loops.

6 Simulation Results

6.1 Visualization

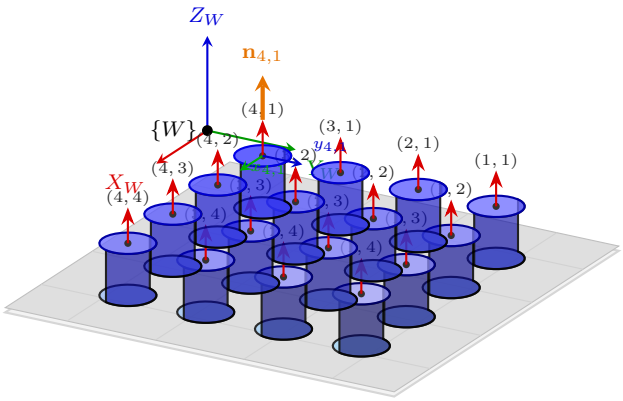


Figure 2: In-perspective volumetric 3D representation of the 4×4 sub-coil grid array. Each cylinder represents an individual sub-coil with low transparency (65%), orientation unit vector $\mathbf{n}_{i,j}$, and depth-based shading (foreground coils shaded darker for enhanced spatial visualization).

6.2 Performance Tables

Table 1: Performance comparison across nominal cluster polar tilt angles θ_0.

Tilt Configuration (θ_0)	**Center $\|B\|_{\text{rms}}$**	**Mean $\mathbf{n}_{\text{rot}}$ Spread**	**Max $\mathbf{n}_{\text{rot}}$ Spread**	**Max $\|B\|$ Noise**
Outward Tilt ($+30.0°$)	1.595 mT	0.9348°	2.8114°	4.1701%
Outward Tilt ($+15.0°$)	3.3465 mT	0.5203°	0.7376°	3.3569%
Flat Baseline ($0.0°$)	4.2846 mT	0.4128°	0.5846°	2.3216%
Optimal Inward ($\theta_0 = -10.0°$)	4.4294 **mT**	0.3821°	0.5745°	1.8283%
Optimal Inward ($\theta_0 = -15.0°$)	4.4072 mT	0.3710°	0.5883°	1.7652%
Extreme Inward ($\theta_0 = -30.0°$)	4.1578 mT	0.3352°	0.5967°	2.1985%

Table 2: Comparison of cluster-level tilting vs. individual sub-coil dish focusing.

Architecture	**Center $\|B\|_{\mathbf{rms}}$**	**Mean $\mathbf{n_{rot}}$ Spread**	**Max $\mathbf{n_{rot}}$ Spread**	**Degrees of Freedom**
Outward Cluster ($\theta_0 = +15°$)	3.3465 mT	0.4760°	0.6470°	4
Flat Cluster ($\theta_0 = 0°$)	4.2846 mT	0.4144°	0.5550°	4
Inward Cluster ($\theta_0 = -15°$)	4.4072 **mT**	0.3710°	0.5883°	**4**
Individual Sub-Coil Dish	5.4295 **mT**	0.4121°	0.9396°	16